Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/24/2018
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	xmao@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8373
Organization Name:	Loma Linda University		
PI Address 1:	Radiation Medicine		
PI Address 2:	11175 Campus Street, Rm A1010, Chan Shun Pavilion		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	12/14/2017	**End Date:**	11/30/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	80NSSC18K0310		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent. The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Task Progress:	Archived ocular tissues from irradiated mice and rabbits have been received from collaborators. We are still waiting on rat tissues from the collaborator. Fixed ocular samples are sectioned for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function. Frozen retina samples are prepared for genomics and proteomics analysis. Our preliminary data showed that space radiation conditions induces acute and long-term response in the retina. Mouse study demonstrates that ionizing radiation induces early apoptosis in mouse retina at a dose as low as 10 cGy. Exposure to low-dose space radiation induces oxidative stress associated apoptosis and changes in rabbit retina and retinal endothelial cells. Proteomic analysis showed that many key pathways responsible for cell death, cell cycle, mitochondrial function, and metabolic stress were significantly altered in irradiated rabbit retina compared to age matched controls. There were significant differences in the responses among radiation types. Our data provide evidence that these cellular responses may affect visual acuity and impact the potential risk of developing late retinal degeneration. The data were presented at the Conference on Normal Tissue Radiation Effects and Countermeasure 2018 and NASA Human Research Program Workshop, 2018. Mice data was also published at Rad. Res. 2018 (PMID: 29741442).
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Pecaut MJ, Hartman R, Nishiyama NC, Bellone J, Boerma M, Nelson G. "Effects of simulated microgravity and/or low-dose radiation on neurovascular remodeling in the mouse brain and eye." Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. , Jul-2018
Abstracts for Journals and Proceedings	Mao XW. "Effects of Low-dose Radiation on Neurovascular Remodeling." Presented at the Conference on Normal Tissue Radiation Effects and Countermeasure (CONTREC), Morrilton, AR, May 14-17, 2018. Conference on Normal Tissue Radiation Effects and Countermeasure (CONTREC), Morrilton, AR, May 14-17, 2018. , May-2018
Abstracts for Journals and Proceedings	Mao XW, Nelson G, Jones T, Campbell-Beacher MS, Nishiyama N, Rodriguez D, Ortloff L, Sun S, Sridharan V, Boerma M, Hauer-Jensen M. "Cardiac and vascular effects of space radiation - CSRR acute risk studies." Presented at 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Articles in Peer-reviewed Journals	Mao XW, Boerma M, Rodriguez D, Campbell-Beachler M, Jones T, Stanbouly S, Sridharan V, Wroe A, Nelson GA. "Acute effect of low-dose space radiation on mouse retina and retinal endothelial cells." Radiat Res. 2018 Jul;190(1):45-52. Epub 2018 May 9. https://doi.org/10.1667/RR14977.1 ; PubMed PMID: 29741442 , Jul-2018